

Audit Open Math Final

OpenMath 2026 · Expanded 27-family judging review

7 October 2026 · Prepared for joint review and signature

Judges: Alejandro Zarzuelo Urdiales and Stephen Wolfram. The existing nineteen award decisions are retained. At the chair's instruction, Robert Huynh is included as a Non-company entrant under the same scoring methodology. Eight additional families extend the reviewed catalogue to twenty-seven; four have positive progress proposals and four have explicit scoring holds. Both signature fields remain blank.

Results and placements

Class / place by S	Contributor / team	S · total	A · best family
Company · 1	Sakana AI	1661.14267	308.025
Non-company · 1	HTPeo	52.6704	49.4214
Non-company · 2	Luke Van Seters / Madhan Jothimani	35.28	35.28
Non-company · 3	Matt Bowring	33.9488	33.9488
Non-company · 4*	Robert Huynh	31.84385	17.064675
Non-company · 5	Chandragupt Sharma	26.2088	26.2088

27 reviewed families · 23 positive-score rows · proposed total 1841.09452 points. This combines the retained 1809.25067 points with Robert's proposed 31.84385. The four held Robert rows are included in the catalogue and excluded from S and A. *Robert's new score and placement await adoption of the stated profile, target and novelty judgments; the original nineteen decisions are unchanged.

How the points express progress

The Open Problem Difficulty Profile (OPDP), developed by Alejandro Zarzuelo Urdiales, and its event extensions provide the common judging framework. $F = m \times p \times D^2 / 1000$. $p=1$ completes the accepted target; **0.15** recognises substantial structural or infinite-subcase progress; **0.05** a quantitative or intermediate advance; **0.01** a limited construction or obstruction. The bonus factor is one. DMS $p=0.15$, the original registered difficulties and the original m values are retained.

S sums distinct contributions; A is the largest family score. Variants, alternative proofs and supporting certificates strengthen their family without another award. Robert is fourth among Non-company entrants by proposed S and fifth by A. New parent-linked children use $m=0.5$; the direct finite programme uses $m=1$. Robert's D values are provisional post-event reference profiles, not retrospectively registered difficulties. The broader-family calibration is explained on page 4.

Judged and evaluated by Alejandro Zarzuelo Urdiales. OpenMath 2026 ran 27 September-2 October: organizer records identify Harvard Innovation Labs for the physical event and a Harvard/MIT programme with a hybrid MIT kickoff. This records event provenance, not university endorsement. <https://rsihouse.ai/openmath> ; <https://luma.com/yzp9abvr>

The progress factors involve judgment. We recognise and commend all participants: their accomplishments matter more than the exact scores.

Why these results matter

OpenMath's strongest achievement is a collection of precise, inspectable mathematical advances. Some settle the exact question asked; others change the best construction, extend a theorem to an infinite class, or reveal why a promising method cannot work. A modest-looking numerical improvement can be difficult to obtain, while a compact theorem can expose a reusable idea. The scoring distinguishes these achievements without treating every partial advance as a solved parent conjecture.

Sakana AI: breadth with complete mathematical stories

Rachel Teo, Wenyi Wang and Hamdi Abderrahmene contribute fourteen awarded families. The subset-product theorem overturns an intuitive expectation: products using all available interval factors still leave arbitrarily long prime-exponent holes. Its $p=1$ and 277.729 points reflect a complete answer, not the numerical size of the constructed examples. The +3 covering theorem receives 308.025 points because it proves the sharp answer to a pre-existing exact subquestion. Other complete targets concern prime-reversal cycles, a sharp symmetry gap, a quartic-power family and the likelihood landscape of a small neural-network model. The Ferrers hierarchy, three-prime exclusion and uniform arithmetic bounds supply structural and quantitative progress on broader questions. Together they show varied mathematical reasoning: carefully designed constructions, inequalities, exact classifications and obstructions. The packet's many supporting dossiers are not counted as that many solved open problems.

HTPeo: local constructions with global consequences

Woohyuk Kang's colouring work turns periodic blocks and repaired seams into constructions valid for infinitely many graph sizes. Small local checks control forbidden patterns throughout the graph, making the method more valuable than a catalogue of examples. The retained $p=0.15$ recognises those families and reductions while broader DMS premises remain open. His exact arrangement of 39 straight lines gives 471 triangular faces: a compact, independently reproducible improvement. The two families place HTPeo first in the Non-company class.

Three further advances in construction and optimisation

Luke Van Seters and Madhan Jothimani combine Clebsch-graph structure, exact density calculations and a lifting argument for Ramsey colourings. Their accepted 35.28 points reward the construction and refinement. Matt Bowring's accepted 33.9488 points recognise a generic matrix-pivot existence theorem: continuity guarantees a phase-balancing choice that may support later quantum-decomposition work. The theorem's value stands independently of the unfinished universal circuit claim. Chandragupt Sharma's exact 3×3 multiplication scheme uses 138 nonzero coefficients at 23 products. Its 26.2088 points recognise a sparsity improvement rather than a product-count reduction. It also clarifies the relationship between a successful new scheme and Sakana's one-factor obstruction: improvements can require coordinated changes to several parts. These distinct contributions give both constructive examples and guidance for future searches.

Evidence and wider contributions

The audit uses statement fidelity, proof closure, exact independent checks and prior-work comparison to decide what each contribution establishes. The archive separates fresh kernel replay, native-evaluation trust, earlier receipts and incomplete checks. This makes the collection useful beyond the event: researchers can inspect a claim, reproduce its certificate and identify the remaining mathematical question. Most awarded Sakana families have fresh scoped replay and endpoint checks; A000224 retains its documented 167/171-source qualification and three unrun selected audits. DMS has a body-identical 228-source route with 123 selected endpoint checks. Chandragupt's seven modules, twelve endpoint audits and 729 tensor identities pass. Luke/Madhan's 51-source packet has 35 endpoint checks, including 23 with native-evaluation trust; Matt's generic theorem has 26-source replay and six standard-axiom checks. These are archived verification findings, with their stated scopes preserved.

Other contributions and reusable work

Established theorems in reusable form. Sakana's ten selected families cover Bernoulli denominators, quaternion scalar equations, an lcm recurrence, Apéry-polynomial irreducibility, reverse-and-add carries, theta cancellation, reciprocal-square arithmetic, determinant integrality, divisor placement and polygon distances. HTPeo's known-theorem work covers perfect matchings, powerful numbers, Sidon bounds, powers-of-2-and-3 completeness, chromatic restrictions, multiplicative closure, Littlewood-Offord behaviour, binomial gcds, polynomial representations, and smaller arithmetic/graph helpers.

Proof infrastructure. Leanification receives a special mention for the hypergraph-container formalization led by Luca Seiki Pereira Fujii. Leon Koerbs / Cameron Fen, Wilson Wu and Jamie Steeg contribute dyadic equivalences, progression inequalities, known three-term results and reusable conclusion interfaces. These libraries make established reasoning available for later machine-checked proofs.

Exact certificates and reconstructions. Collatz descent packets from Chaewon Yoon, HTPeo and Jamie Steeg; finite Busy Beaver runs; CHSH witnesses; known matrix schemes; and Kobon baseline arrangements demonstrate checkable computational work. Older Sakana/HTPeo Ramsey templates and Sakana's restricted Grothendieck certificates are preserved without new frontier points. Rohith Poola's exactly recounted 61-line/1,190-face arrangement remains outside the tally for straight-line priority comparison; his 39/470 witness matches a known construction.

Partial research programmes. Raj Harshit Srirangam's Kobon reductions, Frederik Smits van Oyen's Heilbronn and Ramsey projects, Qichao Wang's Černý manifests, Hichem Benali's finite cyclic five-term progression inquiry, pre-event Ramsey exploration and Matt's remaining circuit programme retain their precise source/proof boundaries. Collectively, the catalogue provides useful methods, reproducible data and clearly identified next questions. These additional contributions are recognised at their delivered scope; they add no points to the nineteen awarded families.

A formalization of known mathematics can still be a major accomplishment: it turns human reasoning into reusable, machine-checkable components. Exact reconstructions provide reliable benchmarks, and partial reductions identify obstacles. Recognising those forms of work alongside original theorems gives a fuller account of what the participants achieved.

Source record

Original source routes: [Publication archive](#); [Known results and formalizations](#); [Verification register](#); [Complete entrant ledger](#).

The original nineteen decisions are preserved from the exact eight-page audit emailed on 6 October 2026. This 7 October edition integrates Robert under the chair's new instruction. Earlier ledgers retain their historical statuses; the original mathematical scopes, incomplete-check disclosures and wider recognition remain in force. The extension is for joint review, not a record of Stephen Wolfram's prior approval.

Robert Huynh: mechanisms, progress and scoring

Robert's eight families concern Erdős 585: finding two edge-disjoint simple cycles on exactly the same vertex set. The strongest checked mechanisms separate quartic extraction, actual-cycle lifting from repeated neighbourhoods, finite lower constructions and a sharply specified recolouring barrier. **None settles the general extremal problem.** Robert's self-reported affiliation is Harvard Business School, MBA programme; publication affiliation remains for author confirmation.

A conservative calibration under the existing p bands

The proved narrow statements are fully proved. The earlier $p=1$ figures were conditional full-target benchmarks. This edition explicitly adopts a broader-family progress model: the representative claims are valued as intermediate, structural or finite advances around the cycle-pair programme, using their provisional local D profiles as reference values. It does not describe a proof as five or fifteen percent correct. Because this is a post-event calibration, the D/target/modality combination needs joint adoption; the original nineteen registered difficulties are unaffected.

Family	D	m	Credited p	Proposed F	Status
QB	494	0.5	0.05	6.1009	Positive proposal
TWIN	477	0.5	0.15	17.064675	Positive proposal
FINITE	435	1	0.01	1.89225	Positive proposal
REG 5	379	0.5	Held	0	Hold: no credit
SUB	525	0.5	Held	0	Hold: no credit
HAAR	468	0.5	Held	0	Hold: no credit
ONE	521	0.5	0.05	6.786025	Positive proposal
BM	772	0.5	Held	0	Hold: no credit

QB $p=0.05$: a sharper intermediate density threshold extracts quartic structure, with the cycle-splitting gap open. **TWIN $p=0.15$:** an infinite restricted host class yields actual cycle pairs through a distinct clone-lifting mechanism. **FINITE $p=0.01$:** only the formal lower witnesses with 31 edges at order 11 and 36 at order 12 are priced; exact upper claims are not upgraded. **ONE $p=0.05$:** an unbounded family establishes a rigorous limitation of the permitted detector method, exceeding a single finite obstruction.

Why four holds: REG 5 needs a specific new witness/minimum-order delta; HAAR needs the exact odd-prime/composite priority comparison; SUB's general substitution theorem and BM's Hamiltonicity adaptation lack matching main Lean endpoints. Supporting formal lemmas do not close those main proofs. Held means unpriced at this review boundary, not incorrect or uninteresting. No private exponent-11 or B2 claim is counted.

The completed public audit compiled 106 authored Lean sources and audited 84 selected declarations with standard kernel axioms only. Ordinary proofs, historical censuses and the written two-round sharpness remain separately labelled. New difficulty estimates retain their C1/C2 editorial limits. Parent CFSD 309 comes from a malformed-import record and is neither inherited nor added. Full proof files and exact source-map links are supplied in the annex and Robert's companion paper.

Annex · Results 1-5

Sakana AI · Rachel Teo, Wenyi Wang and Hamdi Abderrahmene

01 · Subset-product gaps · A060957

277.729 points · $p=1$ · $D=527$ · $m=1$ · [GitHub paper](#)

Choose distinct factors from the complete interval $1, \dots, n$ and multiply them. Attainable products can skip arbitrarily many intermediate powers of a prime, even when products on both sides are attainable. The construction controls prime-exponent patterns so that none of the other interval factors can fill those holes. This answers the actual whole-interval interpolation question. $p=1$: the target is completely resolved; longer gaps and the exact attainable-level classification strengthen the same contribution and receive one award.

02 · Prime-reversal nonperiodicity · A100475

292.681 points · $p=1$ · $D=541$ · $m=1$ · [GitHub paper](#)

Start with a positive integer n , take the n th prime, reverse its decimal digits, and repeat. A step can decrease the value, but one of the next three values always exceeds the starting value. A cycle would have a largest member, so this escape property rules out every positive-start periodic orbit. Uniform growth handles large values; exact checks cover the exceptions. $p=1$: the cycle question is completely answered. The paper records the relevant historical comparisons.

03 · Three-prime exclusion · A000224

58.0326 points · $p=0.15$ · $D=622$ · $m=1$ · [GitHub paper](#)

Count the different square residues modulo n , including zero, and call this number $a(n)$. The result proves that $a(n)(a(n)-1)$ never divides n^2-1 when n is a product of three distinct odd primes. It advances beyond bounded searches by reducing every possible triple to finitely many arithmetic cases, then excluding them using quadratic discriminants. $p=0.15$: this infinite factorization class is completely settled; repeated prime powers and the broader classification remain outside the result.

04 · Ferrers matching hierarchy

63.375 points · $p=0.15$ · $D=650$ · $m=1$ · [GitHub paper](#)

On a staircase-shaped board, a matching chooses cells with no repeated row or column. Under two consecutive optimum-size equalities, any supplied optimal union of r matchings can be preserved while further disjoint matching layers are added, determining subsequent optimum sizes. Tight board corners constrain the available space, and a capacitated matching argument constructs the extensions. The distinctive advance is preservation of every supplied optimum. $p=0.15$: this establishes a substantial hierarchy, while the general Latin Tableau question remains open.

05 · Sharp +3 residual covering bound

308.025 points · $p=1$ · $D=555$ · $m=1$ · [GitHub paper](#)

Consider a family H of r -element sets in which every pair overlaps; $\tau(H)$ is the fewest elements needed to meet every set. The result proves $4\tau(H) \leq |H| + r + 3$, improving the residual +4 term to an optimal +3. A sharper packing argument excludes the remaining obstruction, and removing highly shared elements extends the bound to general families. $p=1$: this sharply answers an explicitly posed exact subquestion. Its importance lies in proving the precise limit, alongside the existing broader Erdős 21 theory.

Annex · Results 6-10

Sakana AI · Constructions, exact variations and uniform quantitative bounds

06 · Four-AP-free reciprocal sum

14.4722 points · $p=0.05$ · $D=538$ · $m=1$ · [GitHub paper](#)

An explicit finite set of positive integers contains no nonconstant four-term arithmetic progression, yet has reciprocal sum greater than 4.44, improving the recorded approximately 4.439753 comparison. Starting from a known base-55 digit construction, it adds six compatible residue families. Digit descent prevents forbidden progressions, and grouped rational estimates certify the larger sum. $p=0.05$: this is a concrete quantitative improvement. Arbitrarily large sums and an infinite progression-free set with divergent reciprocal sum are still unresolved.

07 · Quartic-power classification

207.368 points · $p=1$ · $D=644$ · $m=0.5$ · [GitHub paper](#)

For $H_t(z)=(1+|z|^2)^4-t|z|^4$, ask which positive integer powers are sums of squared absolute values of holomorphic polynomials. The classification is exact for $6 \leq t \leq 7$: power 2 and every power at least 4 work, while power 3 has an explicit threshold. Coefficient positivity and Fourier estimates also show that the least working power grows like $\log(1/\delta)/\delta$ as $\delta=8-t$ tends to zero. $p=1$, $m=\frac{1}{2}$: this precise variation is settled, with a sharp boundary-growth story.

08 · Sharp half-turn defect gap

108.9 points · $p=1$ · $D=330$ · $m=1$ · [GitHub paper](#)

Place $2n$ points in an $n \times n$ grid without three lying on a straight line. Rotating by 180° either preserves the set or leaves at least four points without their rotated partners. Every row and column contains two points, fixing the centroid; cancellation of opposite pairs and parity exclude smaller defects. A ten-point 5×5 example achieves four. $p=1$: the exact symmetry gap is proved sharp. Existence of such configurations for every grid size is a separate question.

09 · Uniform $k\sigma(k)$ fibre bound

16.4738 points · $p=0.05$ · $D=574$ · $m=1$ · [GitHub paper](#)

Let $\sigma(k)$ be the sum of k 's positive divisors. The result limits how many different k produce the same value $k\sigma(k)$. For every $\varepsilon>0$, all sufficiently large targets n have count at most $\exp((C+\varepsilon)\log n/\log \log n)$, with C improved from approximately 0.366204 to 0.363270. Solutions are encoded by repeated prime factors, whose patterns are counted more efficiently. $p=0.05$: the improvement holds uniformly; the parent question still asks for an arbitrarily small coefficient.

10 · Uniform two-cube representation bound

14.045 points · $p=0.05$ · $D=530$ · $m=1$ · [GitHub paper](#)

Count ordered nonnegative pairs (x,y) with $x^3+y^3=n$, including pairs containing zero. For every $\varepsilon>0$, the uniform eventual bound improves to $\exp((\log(5/3)+\varepsilon)\log n/\log \log n)$, replacing $\log(7/4)$. Solutions become codes recording modular cube-root choices, and a separation argument limits their number. A common-divisor decomposition extends the estimate beyond primitive or cube-free cases. $p=0.05$: this tightens the bound for all sufficiently large targets; the much stronger polylogarithmic target remains open.

Annex · Results 11-15

11-14: Sakana AI · 15: HTPeo, with Woohyuk Kang as the recorded paper author

11 · Critical-graph size obstruction

25.99205 points · $p=0.05$ · $D=721$ · $m=1$ · [GitHub paper](#)

For $r \geq 1$, consider graphs that need four colours, become three-colourable when any vertex is removed, yet still need four colours after deleting any set of at most r edges. They require at least $\lceil (27r+19)/4 \rceil$ vertices, with a sharper integer condition giving 33 when $r=4$. Neighbourhood patterns behave like separated codewords, restricting the size. $p=0.05$: this strengthens finite size restrictions; construction and the broader existence question remain separate tasks.

12 · RBM parity likelihood threshold

264.2645 points · $p=1$ · $D=727$ · $m=0.5$ · [GitHub paper](#)

In a model with four observed binary variables and one hidden variable, parity data contain a collective correlation that pairwise statistics cannot detect. The result classifies exactly when parameters predicting the uniform distribution form inferior local likelihood maxima, and when nearby changes escape them. Higher-order analysis gives a sharp threshold, including its equality boundary. $p=1$, $m=\frac{1}{2}$: this completes the specified landscape question. The contribution is a precise account of an optimisation obstacle in this small model, rather than general training dynamics.

13 · Diagonal-pullback obstruction

4.54276 points · $p=0.01$ · $D=674$ · $m=1$ · [GitHub paper](#)

On a 13×13 board with wrapping coordinates, seek points avoiding the specified four-point L-pattern. A natural construction selects entire diagonals using a one-dimensional progression-free set. Every construction of this kind has at most 78 points, while an explicit L-free arrangement has 79. The witness shows that the diagonal recipe imposes unnecessary restrictions. $p=0.01$: this is a precise finite obstruction useful for choosing a better strategy; the optimum and the related infinite Erdős problem remain open.

14 · Matrix one-factor rigidity

5.24176 points · $p=0.01$ · $D=724$ · $m=1$ · [GitHub paper](#)

A matrix-multiplication scheme specifies combinations of the first matrix's entries, combinations of the second matrix's entries, and instructions for assembling their products. At three known 23-product, support-139 schemes, fixing any two parts uniquely determines the third. Exact linear algebra therefore proves that changing just one part cannot simplify these algorithms. $p=0.01$: this is a useful obstruction to a search strategy. Coordinated changes remain possible, so the result is compatible with the support-138 improvement below.

15 · Star edge colouring · DMS

49.4214 points · $p=0.15$ · $D=574$ · $m=1$ · [GitHub paper](#)

Star edge colouring requires adjacent edges to differ and every four-edge path or cycle to use at least three colours. Repeated patterns with repaired seams give five colours for $GP(n,k)$, $1 \leq k \leq 15$ and $n > 2k$, except $GP(3,1)$. Finite local checks establish the construction at every admitted size; small-graph classifications and gluing reductions add supporting structure. $p=0.15$: the novel infinite families and reductions constitute substantial progress. Five broader premises and the general six-colour DMS conjecture remain open.

Annex · Results 16-19

16: HTPeo / Woohyuk Kang · 17: Chandragupt · 18: Luke/Madhan · 19: Matt

16 · 39 lines and 471 triangles

3.249 points · $p=0.01$ · $D=570$ · $m=1$ · [GitHub paper](#)

Arrange 39 straight lines to create triangular regions whose interiors are untouched by other lines. The supplied rational construction produces 471 such regions, improving the recorded pre-event construction by one triangle. Exact intersections and face counts make the achievement reproducible, rather than dependent on the appearance of a drawing. $p=0.01$: this rewards a specific construction improvement. The small numerical gain is meaningful in a constrained extremal problem, although the maximum at 39 lines remains unsettled.

17 · Matrix multiplication with support 138

26.2088 points · $p=0.05$ · $D=724$ · $m=1$ · [GitHub paper](#)

Two 3×3 matrices are multiplied using 23 products and only 138 nonzero coefficients across the input combinations and output assembly, improving the located 139-coefficient reference. Coordinated cancellations preserve every required output term; all 729 tensor identities check, and an exact parameter family retains the support. $p=0.05$: the scheme improves sparsity at fixed product count. It provides a new optimisation benchmark, without establishing 22 products, minimality of 138, or faster practical execution.

18 · Clebsch-based Ramsey construction

35.28 points · $p=0.05$ · $D=840$ · $m=1$ · [GitHub paper](#)

Colour complete-graph edges red or blue while making monochromatic four-vertex cliques scarce. Clebsch-graph structure organises a 192-class probability model and a 3,840-class refinement with density approximately 0.0301388875665. Exact recounts confirm the densities, and averaging transfers the model to actual colourings at the admitted sizes. $p=0.05$: the construction and refinement earn quantitative credit. The achievement combines a concrete bound with a reusable lifting argument; the optimal density and historical first-record comparisons are separate questions.

19 · Generic quantum-pivot theorem

33.9488 points · $p=0.05$ · $D=824$ · $m=1$ · [GitHub paper](#)

Quantum matrix decompositions use rotation angles encoding phase differences. For the specified generic matrices, a suitable one-qubit pivot makes one selected phase-balancing angle exactly zero. A first rotation avoids singularities; continuous phase tracking turns opposite endpoint imbalances into an exact zero through the intermediate value theorem. $p=0.05$: this establishes an intermediate existence theorem relevant to simplifying decomposition. The circuit-construction bridge, nongeneric cases and universal $21/48$ CNOT-count claim remain unproved.

Annex · Results 20-22

Robert Huynh · Original mathematics and Lean development; Alejandro Zarzuelo Urdiales · Research synthesis, exposition and evaluation. Coauthor manuscript prepared for author review.

20 · Quartic extraction at maximum degree six

6.1009 proposed points · $p=0.05$ · $D=494$ · $m=0.5$

A finite simple bipartite graph on $n \geq 3$ vertices with maximum degree at most six and at least $3n-5$ edges always contains a nonempty 4-regular subgraph. A minimal-counterexample argument turns degree deficits into tightly constrained cuts and ports. Covering the possible seven-edge cut and combining the resulting hub and petal constraints forces the required subgraph. The subgraph need not span the host or split into two Hamilton cycles. QB4 and the prescribed-vertex quartic variant strengthen this same family; they are not extra results. The unrestricted cycle-pair problem remains open.

$p=0.05$ prices the extraction-threshold advance; an actual cycle-pair bridge remains open.

Exact source routes: qb5; qb4; exists_four_regular_avoiding_of_bipartite_six_regular.

21 · Actual cycle pairs from repeated neighbourhoods

17.064675 proposed points · $p=0.15$ · $D=477$ · $m=0.5$

In a finite simple bipartite graph with maximum degree six and exactly $3v-2$ edges, suppose every left vertex has another distinct left vertex with the same neighbourhood. Then there are two edge-disjoint simple cycles on precisely the same actual vertex set. The edge count balances the two shores and their deficits. A Boolean polynomial selects a quotient with degree zero or four at grouped left vertices and zero or two on the right. An Euler circuit then assigns two actual twin vertices to each group; exchanging their roles gives the second cycle. The common support need not span the graph. Grouped and ungrouped formal statements constitute one family, and the additional six-regular sufficient class is ordinary exposition. The theorem does not cover arbitrary twin-free six-regular hosts.

$p=0.15$ prices substantial infinite-class progress with actual cycles; its exact density and twin hypotheses remain essential.

Exact source routes: hasPair_of_no_singleton_twins; groupedGraph_hasPair_of_no_singletons.

22 · Finite extremal values and explicit lower constructions

1.89225 proposed points · $p=0.01$ · $D=435$ · $m=1$

For $f(n)$, the maximum edge count of an n -vertex simple graph without a cycle pair, two explicit formal lower witnesses give $f(11) \geq 31$ and $f(12) \geq 36$. These are the narrow construction proposals priced in the current assessment. The supplied graph constructions have exact vertex and edge counts and faithful formal pair-freeness proofs. Deletion bounds and finite catalogue checks provide supporting context; the two lower witnesses are consolidated as one contribution. The wider report lists exact values through twelve, but the formal exact-value boundary is through seven and supplied computations were freshly replayed only through ten. Upper exclusions at eleven and twelve remain source-reported. A lower witness does not establish an exact value, and the linear constructions do not improve the known asymptotic lower order.

$p=0.01$ prices the two new formal lower witnesses. Values through seven may overlap known mathematics; orders 11/12 upper exclusions and large censuses are historical.

Exact source routes: thirty_one_le_maxEdges_eleven; thirty_six_le_maxEdges_twelve.

Annex · Results 23-25

Robert Huynh · Original mathematics and Lean development; Alejandro Zarzuelo Urdiales · Research synthesis, exposition and evaluation. Coauthor manuscript prepared for author review.

23 · Pair-free five-regular witnesses

Held · 0 credited points · p unassigned pending the stated gate · D=379 · m=0.5

Explicit simple five-regular pair-free graphs exist on 18 and 32 vertices, with a bipartite example on 104 vertices. The refined paper also gives a separately checked alternative 104-vertex edge set. Any cut crossed by a cycle pair has at least four edges. Smaller cuts confine both cycles to a block; successive deletion of vertices of degree at most three then excludes a pair inside that block. This provides short exact certificates for the constructions. The 18-vertex witness alone does not prove that 18 is the minimum possible order: that needs the full exclusion census through 16. Neither 104-vertex example is claimed to have minimum bipartite order. Degree-five obstructions do not refute the degree-six conjecture.

Hold: correctness of the witnesses is checked, but a distinct new mathematical delta and the minimum-order claim need separate evidence.

Exact source routes: exists_five_regular_pairfree; exists_bipartite_five_regular_pairfree; exists_five_regular_pairfree.

24 · Substitution and the prescribed-vertex reduction

Held · 0 credited points · p unassigned pending the stated gate · D=525 · m=0.5

For $d \leq 7$, a valid simple substitution of pair-free bipartite d -gadgets into a pair-free d -regular loopless multigraph remains pair-free. The mechanism also relates the six-regular bipartite assertion to its prescribed-vertex deletion form. Counting the shores of a hypothetical pair shows that each visited gadget uses zero or exactly four link edges. Each cycle therefore crosses a gadget in a single passage, projecting to two genuine edge-disjoint cycles in the skeleton. Parallel-edge identities are retained. The full substitution and equivalence theorem is an ordinary written argument; selected small-cut and splitting lemmas are narrower formal support. Arbitrary port assignments need not give a simple graph, and equivalence of two open assertions proves neither. The 104-vertex witness is scored in REG 5 only.

Hold: the general projection/substitution theorem is an ordinary proof; formal cut and balance helpers are support only.

Exact source routes: Support: splitGraph_pairfree; Support: not_hasPairF_of_small_cut; Ordinary full proof.

25 · Prime-field Haar cycle pairs and cyclic certificates

Held · 0 credited points · p unassigned pending the stated gate · D=468 · m=0.5

For any prime-field module V and finite palette S with at least four distinct shifts, the bipartite Haar graph on two labelled copies of V contains a finite actual cycle pair. Separate cyclic templates on $\mathbb{Z}/(n^2)$ with shifts $0, n, 2n, 1$ have pairs for $n \geq 3$. Translate four colours to include zero and work in their finite affine span. Rank-two coordinate sweeps give complementary full orbits; rank three joins level cycles by exchanging marked matching edges. Odd and even cyclic certificates have separately checked orbit and connectedness obligations. The broad forward theorem allows an infinite ambient V , whereas the selected formal iff assumes finite ambient type. Prime-line and characteristic-two cases are known background. The cyclic template is not a classification of all composite-group Haar graphs, and the precise higher-rank priority delta remains held.

Hold: prime-line and characteristic-two cases overlap located prior mechanisms; the odd-prime rank-two/rank-three and composite novelty deltas need comparison.

Exact source routes: rankThree_hasPair; hasPair_of_four; hasTwoEdgeDisjointCyclesSameVertexSet_all; hasPair_iff; hasPair.

Annex · Results 26-27

Robert Huynh · Original mathematics and Lean development; Alejandro Zarzuelo Urdiales · Research synthesis, exposition and evaluation. Coauthor manuscript prepared for author review.

26 · A one-round method barrier with written two-round sharpness

6.786025 proposed points · $p=0.05$ · $D=521$ · $m=0.5$

On canonical characteristic-three parabola hosts, one allowed recolouring round may swap any union of whole components of a two-colour factor. No cycle exposed this way has an edge-disjoint partner on the same actual support, even at degrees exceeding every fixed polylogarithmic threshold. Up to four distinct palette vectors are affinely independent over F_3 . A one-round exposed cycle uses at most three original colours; its induced support has degree at most three, preventing a pair. A rank-three Haar construction then supplies an ordinary two-round witness on 54 vertices. These hosts already contain pairs. The barrier concerns the supplied colouring and allowed detector, not unrestricted pair search. Exactly two component-union rounds is an ordinary editorial consequence; no new formal composition endpoint or minimum number of individual component flips is claimed.

$p=0.05$ prices the infinite method limitation. Classical hosts can contain ordinary pairs; the two-round composition is written evidence and earns no second score.

Exact source routes: [arbitrary_polylog_barrier](#).

27 · Quantitative bipartite Hamiltonicity and resilience

Held · 0 credited points · p unassigned pending the stated gate · $D=772$ · $m=0.5$

For positive-degree regular bipartite graphs on $n \geq 4$ vertices, the written adaptation proposes a Hamiltonicity threshold at degree $C\delta^{-5}(\log n)^3$ under a one-sided spectral gap. It retains a conservative δ^{-6} branch, corresponding to γ^{-10} versus γ^{-12} in expansion notation. Independent sampling from the two shores preserves the appropriate singular-value gap. A side-respecting router and matching argument assembles genuine host paths. A spanning-factor lemma and balanced vertex deletion give two-cycle and resilience consequences with explicit normalizations. The main BM, factor and deletion results are written, without a selected Lean endpoint. Bipartite Hamiltonicity existence was already announced in prior work; only a checked quantitative refinement can be new. The missing extraction premise E110 and unavailable private exponent-eleven claims yield no unconditional bound for Erdős 585.

Hold: BM/F/V110 have no main Lean endpoint. The quantitative power-five refinement and imported ingredients need specialist review; the bipartite analogue itself was already announced.

Exact source routes: [Written proof 1](#); [Written proof 2](#); [Written proof 3](#).

OpenMath 2026

Mathematical advances, shared foundations

Twenty-seven reviewed families bring together unexpected integer-product gaps, sharper combinatorial limits, neural-network likelihood structure, sparse matrix multiplication, a quantum-decomposition lemma and structured advances around the cycle-pair problem. Zarzuelo's OPDP and its event extensions provide a common way to explain their different kinds of progress.

Sakana AI retains first place in the Company class. By S, the extended Non-company order is HTPeo, Luke Van Seters / Madhan Jothimani, Matt Bowring, Robert Huynh and Chandragupt Sharma. Robert's inclusion and reference-profile calibration are proposed for joint adoption; his largest-family A places fifth in that class.

Contributor / team	Class / place by S	Positive / reviewed	Points
Sakana AI · Rachel Teo / Wenyi Wang / Hamdi Abderrahmene	Company · 1	14 / 14	1661.14267
HTPeo · Woohyuk Kang / Hyunjin Lee / Sanghyeon Lee / Youchan Oh	Non-company · 1	2 / 2	52.6704
Luke Van Seters / Madhan Jothimani	Non-company · 2	1 / 1	35.28
Matt Bowring	Non-company · 3	1 / 1	33.9488
Robert Huynh	Non-company · 4*	4 / 8	31.84385
Chandragupt Sharma	Non-company · 5	1 / 1	26.2088

Proposed combined total: 1841.09452 points. Nineteen original awards remain intact; four added positive progress proposals and four added held families complete the 27-row review. No held row contributes to S or A.

An achievement beyond the originality tally

Leanification retains its special mention for the container-theorem formalization. The known-theorem libraries, progression infrastructure, exact certificates and scoped research programmes remain recognised. Robert's unpriced ordinary proofs, structured classifications and verified witness work are also useful publication and research material at their stated scope.

The progress factors involve judgment; the mathematical accomplishments matter more than exact point totals. This unsigned edition records the chair's expanded review and requests the other judge's assessment. By signing below, the judges confirm the scopes, adopted scores and recognition in this edition.

Alejandro Zarzuelo Urdiales

Judge · Date: _____

Stephen Wolfram

Judge · Date: _____

Review routes: [Central judging archive](#); [Robert repository](#) and written papers.